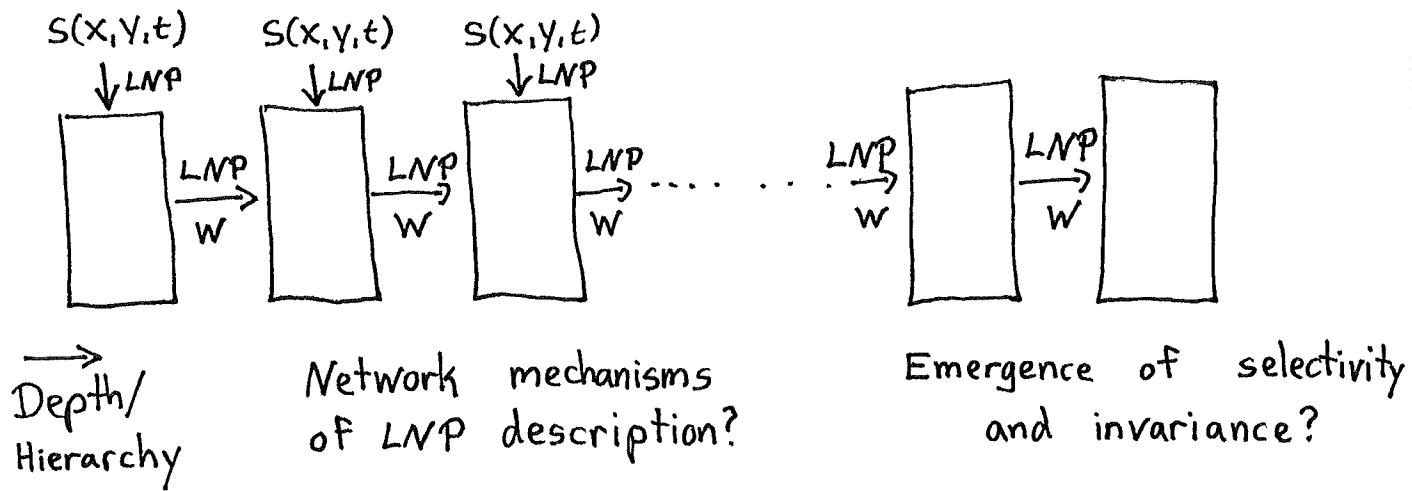
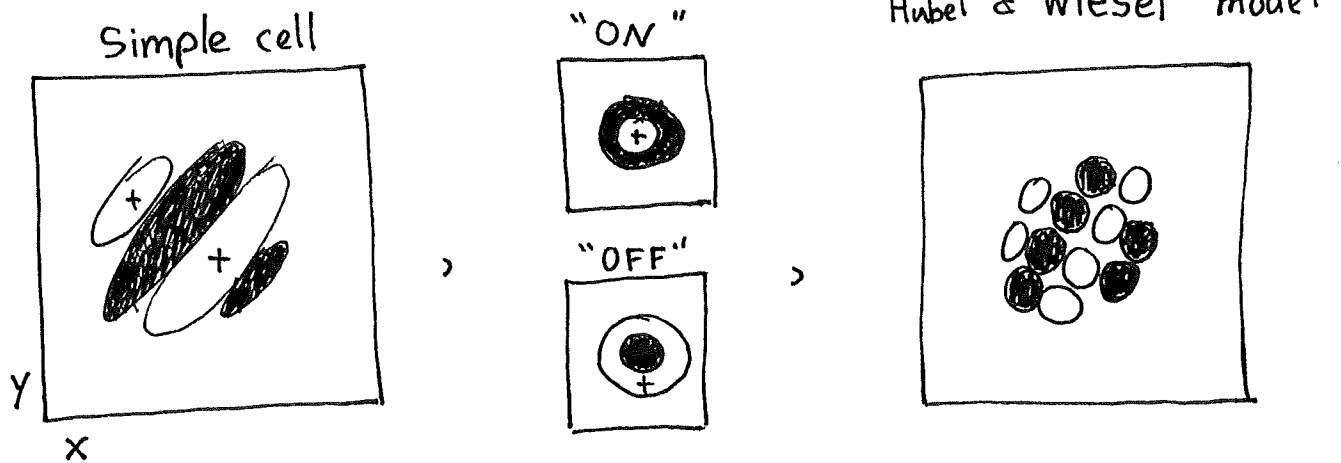


LNP cascades:

LNP models can include neural network interactions.



Simple cell example:



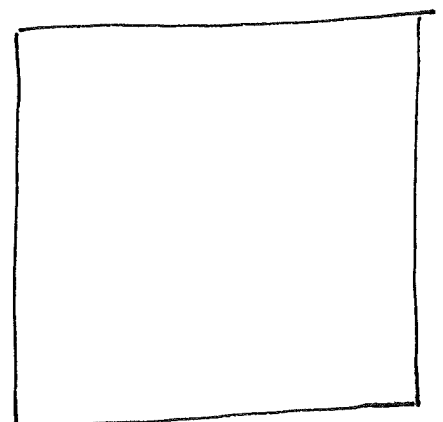
Emergence of edge selectivity.

V1 Complex cell:
Stimulus Response



.....
.....
.....
.....

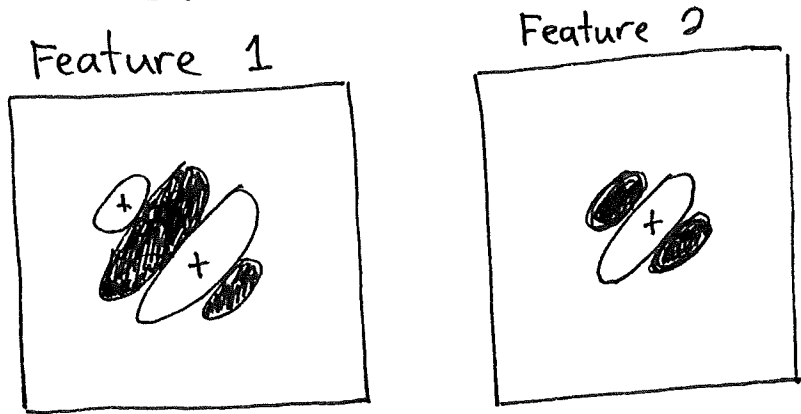
Linear receptive field



$\Delta = 0$

Null spike-triggered average

Spike-triggered covariance works.



2D nonlinearity:

$$h(t) = (L_1 * s)^2 + (L_2 * s)^2$$

"Energy model"

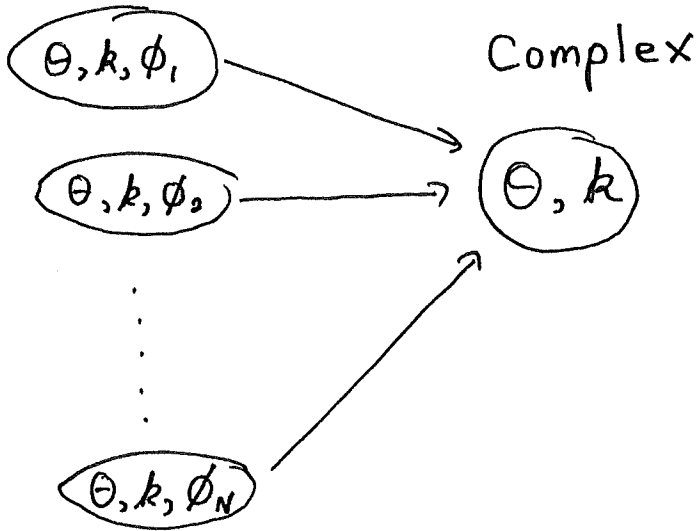
"Quadrature pair"

Simple cell interpretation of energy model

$$h(t) = [L_1 * s]_+^2 + [(L_1) * s]_+^2 + [L_2 * s]_+^2 + [(L_2) * s]_+^2$$

Hubel and Wiesel model

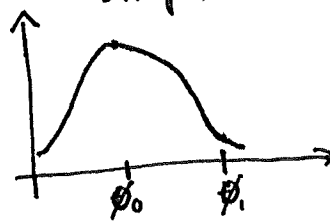
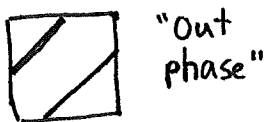
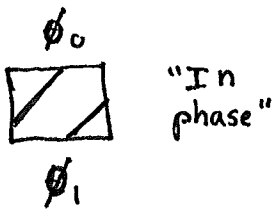
Simple



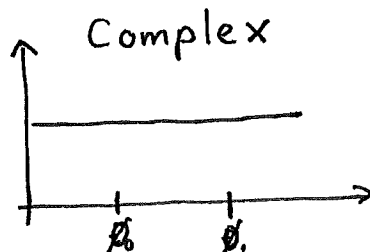
Complex

Emergence of phase invariance:

Stimuli



responds most at "In phase"



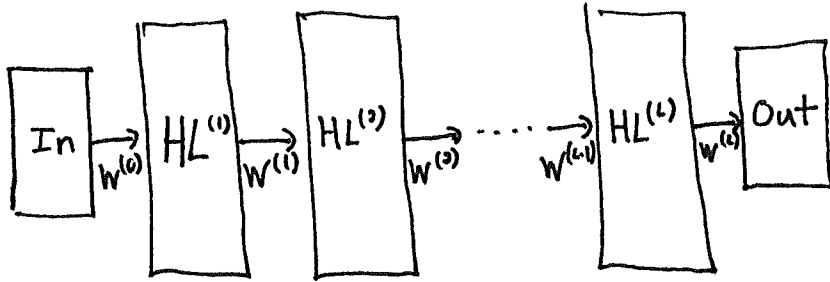
equal response at all phases
($\cos^2 \phi + \sin^2 \phi = 1$)

Performance-optimized models

IT is deep in the hierarchy, highly selective for objects, and highly invariant to details.

How can we model such complex neural responses?

Step 1: Build a model that ^{can} classify objects well



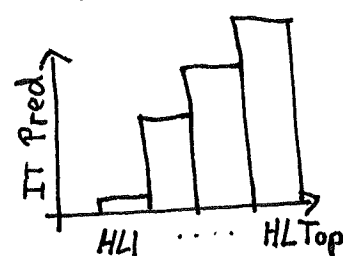
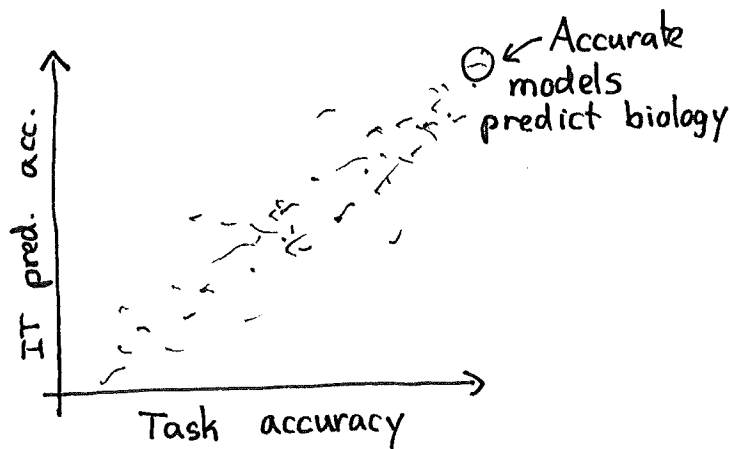
$$\{\hat{W}^{(0)}, \hat{W}^{(1)}, \dots, \hat{W}^{(L)}\} = \arg \min_{\{W^{(0)}, \dots, W^{(L)}\}} \mathcal{L}(W^{(0)}, \dots, W^{(L)})$$

optimize with machine learning
Quantifies object recognition performance (classification)

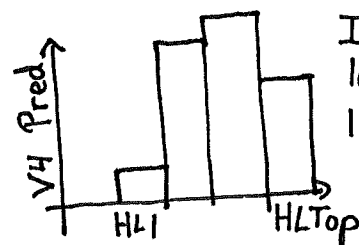
Step 2: Model biological neurons with artificial ones

$$\underbrace{\hat{h}_i}_{\text{Bio}} = \sum_{j=1}^N \underbrace{\beta_{ij}}_{\text{weighting coefficients}} \underbrace{Z_j}_{\text{Artificial}} ; \quad \hat{\beta} = \arg \min_{\beta} \underbrace{\sum_m (h_{im} - \hat{h}_{im})^2 / P}_{\text{Mean squared error}}$$

Paradigm pioneered by Yamins and DiCarlo and produces state of the art models.



Deep layers look like IT

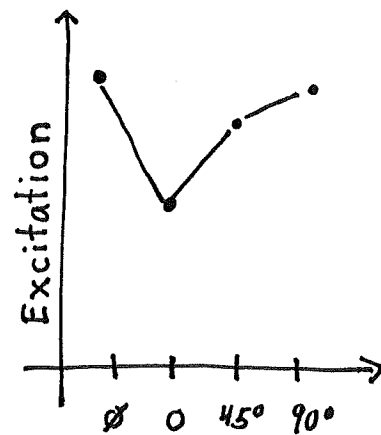
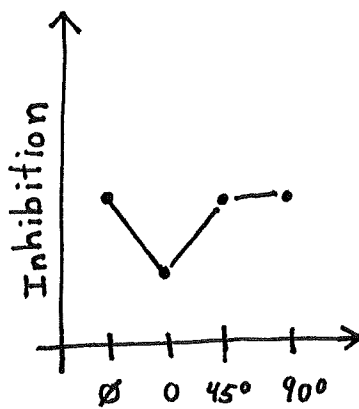
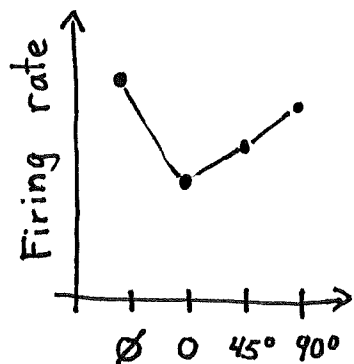
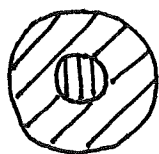


Intermediate layers look like V4.

Recurrence in vision:

We've so far emphasized the power of deep, feedforward processing. Visual neurons also interconnect within a region, and this has important implications for neural coding.

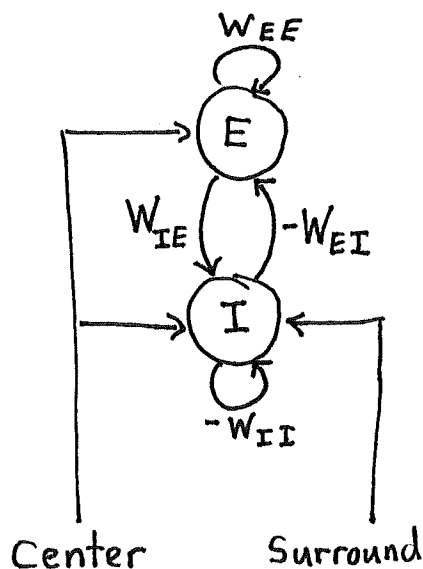
Surround suppression:



Surround orientation

The surround decreases the firing rate, which naively suggests that the surround ~~in~~ inhibits the cell. In 2009, David Ferster's lab at Northwestern surprisingly found that the surround decreases the inhibition. The firing rate goes down because excitation decreases more.

Recurrent neural network model:



$$\tau_E \frac{d\lambda_E}{dt} = -\lambda_E + \sum_E (W_{EE} \lambda_E - W_{EI} \lambda_I + I_E)$$

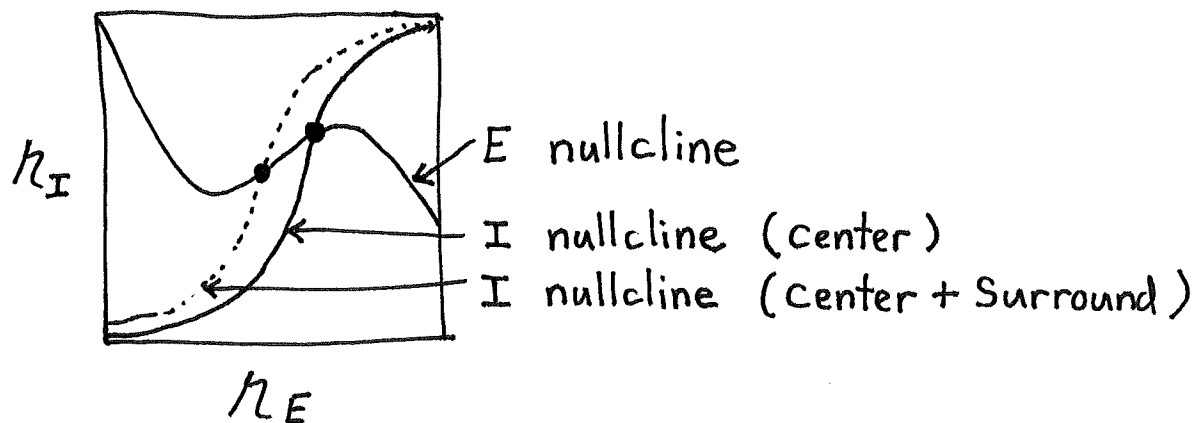
$$\tau_I \frac{d\lambda_I}{dt} = -\lambda_I + \sum_I (-W_{II} \lambda_I + W_{IE} \lambda_E + I_I)$$

The firing rates settle at values such that both derivatives equal zero.

$$\frac{d\lambda_E}{dt} = 0 \Rightarrow \lambda_E = \mathcal{F}_E(W_{EE}\lambda_E - W_{EI}\lambda_I + I_E)$$

$$\frac{d\lambda_I}{dt} = 0 \Rightarrow \lambda_I = \mathcal{F}_I(W_{IE}\lambda_E - W_{II}\lambda_I + I_I)$$

These define two curves in (λ_E, λ_I) space, termed nullclines, whose intersections determine possible steady states of the network. In the same 2009 paper, Ken Miller's lab proposed an explanation for the Ferster experimental result.



The surround input increases I_I , shifting the I nullcline upwards. Both λ_E and λ_I are decreased in the new steady state, thereby decreasing both excitation and inhibition to the excitatory cells.

The nullcline geometry above indicates an "inhibition stabilized network." We need to better understand neural network dynamics to appreciate the name.